If Newton's conception of absolute space and absolute time were correct, everyone would agree on a single slicing of spacetime. Each slice would represent absolute space as viewed at a given moment of absolute time. This, however, is not how the world works, and the shift from rigid Newtonian time to the newfound Einsteinian flexibility inspires a shift in our metaphor. Rather than viewing spacetime as a rigid flip book, it will sometimes be useful to think of it as a huge, fresh loaf of bread. In place of the fixed pages that make up a book—the fixed Newtonian time slices— think of the variety of angles at which you can slice a loaf into parallel pieces of bread, as in Figure 3.5a. Each piece of bread represents space at one moment of time from one observer's perspective. But as illustrated in Figure 3.5b, another observer, moving relative to the first, will slice the spacetime loaf at a different angle. The greater the relative velocity of the two observers, the larger the angle between their respective parallel slices (as explained in the endnotes, the speed limit set by light translates into a maximum 45° rotation angle for these slicings[9]) and the greater the discrepancy between what the observers will report as having happened at the same moment.

The Bucket, According to Special Relativity

The relativity of time and space requires a dramatic change in our thinking. Yet there is an important point, mentioned earlier and illustrated now by the loaf of bread, which often gets lost: *not everything in relativity is relative.*Even if you and I were to imagine slicing up a loaf of bread in two different ways, there is still something that we would fully agree upon: the totality of the loaf itself. Although our slices would differ, if I were to imagine putting all of my slices together and you were to imagine doing the same for all of your slices, we would reconstitute the same loaf of bread. How could it be otherwise? We both imagined cutting up the same loaf.

Similarly, the totality of all the slices of space at successive moments of time, from any single observer's perspective (see Figure 3.4), collectively yield the same region of spacetime. Different observers slice up a region of spacetime in different ways, but the region itself, like the loaf of bread, has an independent existence. Thus, although Newton definitely got it wrong, his intuition that there was something absolute,